Not All Instructions Are Forgotten Equal: Per-Instruction Retention Is a Model-by-Instruction Interaction

Tomás P. Korenblit

Universidad Nacional de San Martín, Buenos Aires, Argentina
tomaskorenblit@gmail.com

Abstract. Large language models lose adherence to user-stated preferences over long conversations, and the common defense, restating every instruction each turn, is expensive and indiscriminate. Whether a selective alternative is possible depends on how retention is structured across instructions and models. We measure per-instruction compliance in realistic multi-turn coding sessions across five models, twelve instruction types, and three open-source Python codebases, graded by deterministic checkers, and we compare a ladder of Bayesian ordered-logistic models. The decisive structure is neither a per-instruction main effect nor a per-model main effect but their *interaction*: a model that adds a model-by-instruction interaction term improves out-of-sample fit by about 59 ELPD (five standard errors) over additive models, while additive per-model terms add nothing. Which instruction is retained depends on the model. We also find that the "backfire" reported in the conference version of this study, where stating one instruction lowered compliance, does not survive: it is a measurement artifact, confirmed both by a corrected checker and by a negation-versus-positive framing experiment that finds no effect of polarity. We argue that the recommended model, a hierarchical Bayesian graded-response model with a model-by-instruction interaction, is the appropriate analysis for evaluation data generally, and that reporting raw means or additive effects mischaracterizes what evaluations measure. The finding matters most for agents deployed in long-running loops without human oversight, where a guardrail that decays does so silently.

Keywords: instruction retention , item response theory , Bayesian inference , model evaluation , AI safety

1 Introduction

Language models are increasingly deployed as autonomous agents that run in loops, call tools, and act over hundreds of turns with no human reviewing each step. An agent is governed by its instructions: the operational limits and safety guardrails it is given, usually once and at the start, that must hold for the whole run. Yet adherence to instructions erodes as the context grows, a phenomenon we refer to as *context rot*. Laban et al. (2025) measured average compliance drops of

39% across 15 models in multi-turn settings. The cause is architectural: attention is not uniformly distributed across the context window, and information in middle positions receives less weight than at the boundaries (Chowdhury, 2026; Liu et al., 2024). An instruction stated once and then buried under accumulating tool output competes for attention against everything that follows it, and in a deployed loop there is no user to notice the lapse and restate it.

This is a safety concern before it is a cost concern. When the model drops a preference, a chat user notices and restates it; a deployed agent has no such supervisor, so a guardrail that decays does so silently while the agent keeps acting through tools whose effects are real and often irreversible (Mu et al., 2025). Repeating every instruction each turn, the standard mitigation, multiplies prompt cost and still cannot tell an operator which constraint remains in force. Because safety is conjunctive, an aggregate compliance rate says nothing about whether the one rule that matters is among those that lapsed.

Whether reinforcement can be targeted instead of uniform depends on how retention is structured. The conference version of this work (Korenblit, 2026) fit a per-instruction Bayesian model to a single-model dataset and reported that retention varies widely across instructions, that one instruction was harmed by being stated, and that the model and codebase explained little variance. This paper revisits those claims with a larger, multi-model dataset, a corrected scoring pipeline, a framing experiment, and a wider model comparison. The picture changes.

Our contributions are: (i) we show that per-instruction retention is governed by a *model-by-instruction interaction*, not by a main effect of either factor: an interaction model improves leave-one-out fit by about 59 ELPD over additive models, and additive per-model terms add nothing; (ii) we show that the reported "backfire" is a measurement artifact, with two independent lines of evidence (a corrected checker and a polarity-framing experiment); (iii) we quantify how fragile variance-component conclusions are to the pool of models analyzed, and which conclusions survive; and (iv) we argue, and demonstrate on this dataset, that a hierarchical Bayesian graded-response model with a model-by-instruction interaction is the appropriate default for analyzing evaluation data, situating it against item-response-theory and generalizability-theory practice.

2 Related work

Instruction degradation in long contexts. Several studies establish that adherence falls as conversations grow. Laban et al. (2025) report an average 39% compliance drop across 15 models; He et al. (2024) show multi-instruction accuracy falling within three turns; Du et al. (2025) find that a larger context window does not by itself fix the loss. None asks whether different instructions are retained to different degrees. On mechanism, Liu et al. (2024) document a U-shaped positional attention curve and Mu et al. (2025) a saturation effect. Existing mitigations reinforce uniformly: periodic reminders (Dongre et al., 2025) and prompt repetition (Leviathan et al., 2025), which can be counterproductive for instructions where restating degrades compliance.

Evaluation as measurement, and item response theory. A parallel literature argues that benchmark scores should be treated as measurements with uncertainty rather than raw means. Miller (2024) imports experiment-analysis machinery (standard errors, clustering, paired differences); Bowyer et al. (2025) shows the normal approximation fails at the small sample sizes typical of specialized evals. Construct-validity work (Bean et al., 2025) finds widespread operationalization failures across benchmarks, and generalizability theory has been applied to decompose LLM-scoring variance across facets (Brennan, 2001; Song et al., 2025). Ranking is increasingly treated as estimation, with Bradley–Terry models and confidence intervals (Chiang et al., 2024). Item response theory casts systems as examinees with a latent ability and items as having difficulty and discrimination (De Boeck & Wilson, 2004; Lalor et al., 2016; Samejima, 1969); it has been used to build evaluation scales, reweight leaderboards (Rodriguez et al., 2021), compare test sets (Vania et al., 2021), and build efficient benchmarks (Polo et al., 2024). This work is almost entirely on binary correct/incorrect outcomes and estimated by variational approximation; the one graded-response application we are aware of targets judge reliability, not the systems under test (Choi et al., 2026). Our model sits in the explanatory-IRT tradition (De Boeck & Wilson, 2004) with crossed random effects (Baayen et al., 2008), but treats responses as ordinal, casts the experimental manipulation as a covariate on ability, and estimates the full hierarchy by MCMC.

Tracking what a model has forgotten. Bayesian Knowledge Tracing (Corbett & Anderson, 1994) estimates per-concept mastery from response sequences. We adapt the framing: the model is the student and its instructions are the concepts. A reinforcement scheme would only be worthwhile if retention is heterogeneous and predictable, which is what we test.

3 Method

3.1 Decision planting

We simulate a coding assistant working on a real open-source codebase over 25 turns. Real source files are injected at turns 0–19, building context pressure. In the **treatment** condition, 12 coding preferences are embedded as casual inline requests in the user's messages, six early (turns 1–7) and six late (turns 14–19), spanning code style, architecture, testing, naming, dependencies, and documentation. They are phrased as natural user preferences, not system-prompt rules (e.g., "Important: when writing array operations for this project, always use NumPy broadcasting instead of for loops"). The **baseline** condition is identical but plants nothing, measuring spontaneous compliance against the same probes. At turns 20–25 the user requests code-generation tasks that elicit each decision; each test turn probes two decisions, one early and one late.

To separate an instruction's content from its phrasing, two further arms hold meaning constant and vary only polarity: **positive** and **negation** paraphrases of each flippable instruction. The framing effect is estimated arm against arm, since the canonical treatment wording differs from both by more than polarity.

3.2 Compliance measurement

Each decision is scored by a deterministic checker that parses the generated code with pattern matching, with no LLM in the scoring loop, so results are reproducible across runs. Each checker emits an ordinal 0–3 score (0 = ignored, 3 = fully followed) or abstains (-1, excluded) when the code gives no opportunity to apply the rule. The conference study's `no_new_imports` checker counted any import statement as new; the corrected checker treats re-showing an already-imported module as compliant (§5.3). As a secondary channel, a panel of three cross-family LLM judges scores the same outputs blind to condition and model; no judge shares a family with a candidate, removing the conference study's self-judging bias. We report checker–judge agreement in §5.5.

3.3 A ladder of Bayesian models

We model the ordinal scores with hierarchical ordered-logistic (cumulative-link) regression, comparing five specifications of increasing structure by leave-one-out cross-validation (Vehtari et al., 2017). Let d index decision, m model, k codebase, and let t_i be the treatment indicator. All effects use non-centered parameterizations and offset cutpoints; priors follow standard weakly-informative choices for ordinal models (Bürkner, 2021; McElreath, 2020).

M0 (the conference model) is per-decision only: $\eta_i = \alpha_d + \beta_d\, t_i$, with α_d, β_d given hierarchical priors. **M4** adds additive per-model and per-codebase intercepts and a per-model treatment slope: $\eta_i = \alpha_d + \beta_d t_i + g_m + t_i\, s_m + g_k$. **M_corr** draws (g_m, s_m) from a correlated bivariate normal (LKJ prior). **M_irt** is an explanatory graded-response model: it replaces the additive per-model intercept with a multiplicative discrimination term, $\eta_i = b_d + a_d\,\theta_m + \beta_d t_i + g_k$, where θ_m is a latent model ability and $a_d > 0$ a per-instruction discrimination, the IRT form of a model-by-instruction interaction. **M8** keeps the additive structure of M4 and adds an identifiable interaction *variance component*: a per-cell treatment effect $z_{m,d} \sim \mathcal{N}(0, \sigma_{\text{int}})$, so $\eta_i = \alpha_d + \beta_d t_i + g_m + t_i s_m + g_k + t_i\, z_{m,d}$. The key scientific parameter shifts from σ_β (spread of per-instruction effects) to σ_{int} (spread of the model-by-instruction interaction). A variance-component interaction is preferred to a rank-1 factor model, which is unidentified up to sign and scale; the factor version did not converge ($\hat{R} = 1.74$), while M8 samples cleanly. Fit in PyMC (Abril-Pla et al., 2023) with NUTS via nutpie (Hoffman & Gelman, 2014; Seyboldt, 2024), four chains, 2000 draws each, `target_accept` $= 0.99$.

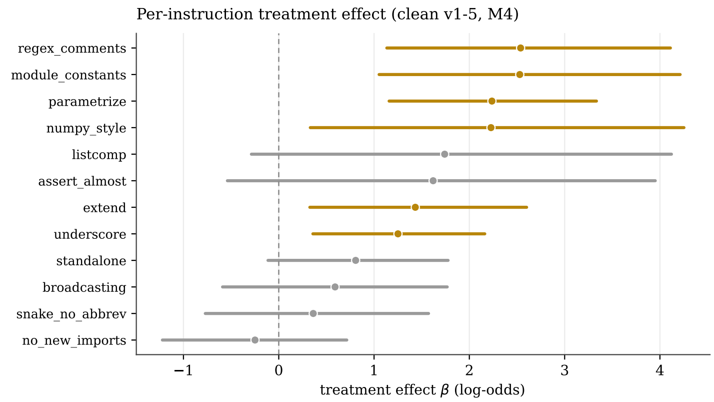

Fig. 1. Per-instruction treatment effect β with 94% HDI (additive model, clean five-model set). Gold: interval above zero; grey: crosses zero. The conference study's -2.36 "backfire" on `dependencies_no_new` is gone (bottom).

4 Experimental setup

Three open-source Python libraries span a range of context pressure: Bambi (median peak ≈85K tokens by the test turns), ArviZ (≈149K), and PyMC (≈187K). Five instruction-following models were evaluated with both baseline and treatment in every codebase: Qwen 3.5 (27B), Gemini 3.1 Flash Lite, GPT-5.4-nano, Nemotron 3 Super (120B), and Gemma 4 (26B). Each condition was run for five epochs to average per-call stochasticity. After excluding abstentions, the primary dataset is 1,067 compliance observations across 5 models, 12 decision types, 3 codebases, and 2 conditions; the framing arms add a further set across three of the models. The pipeline is a port to the Inspect evaluation framework; the conference study used a bespoke runner with a single baseline model and a same-family judge.

5 Results

5.1 Retention is heterogeneous

The per-instruction treatment effect varies substantially: in the additive model its group-level standard deviation is $\sigma_\beta = 1.19$ (94% HDI $[0.55, 1.99]$), excluding zero. Five to seven of the twelve instructions, depending on specification, have a treatment effect whose interval excludes zero (Fig. 1); the strongest are `dependencies_module_constants`, `docs_regex_comments`, `testing_parametrize`, and `docs_numpy_style`. This is about half the magnitude reported by the conference study ($\sigma_\beta = 2.11$); §5.2 explains why.

Table 1. Model comparison by leave-one-out cross-validation (clean five-model set, 1,067 observations). Higher ELPD is better; ΔELPD is the difference from the best model with its standard error.

Model	Interaction structure	ELPD	ΔELPD (SE)
M8	model×instruction variance	−545.9	—
M_irt	2PL discrimination $a_d\theta_m$	−555.4	9.5 (10.5)
M_corr	correlated per-model slopes	−604.4	58.5 (11.2)
M4	additive per-model	−604.8	58.9 (11.2)
M0	per-instruction only	−607.2	61.3 (11.7)

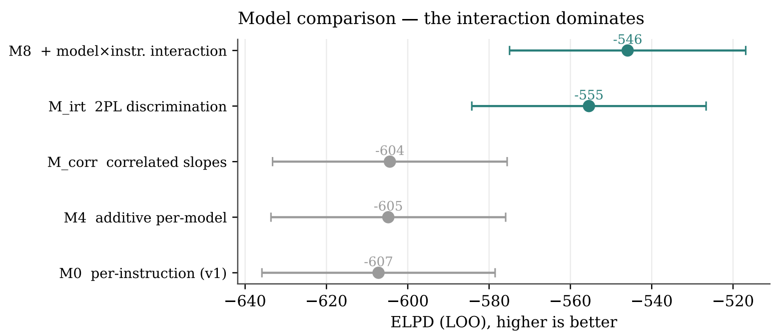

Fig. 2. Out-of-sample fit (ELPD, higher is better) with standard errors. The interaction models (teal) separate cleanly from the additive models (grey), which cluster together.

5.2 The structure is a model-by-instruction interaction

Comparing the model ladder by leave-one-out cross-validation (Table 1, Fig. 2) gives a single clear result. The two models that capture a model-by-instruction interaction, M8 and M_irt, fit decisively better than every additive model, by about 59 ELPD (more than five standard errors). The two interaction models are not distinguishable from each other ($\Delta\text{ELPD} = 9.5 \pm 10.5$), so it is the interaction that matters, not its exact parameterization. Crucially, the additive per-model terms add nothing: M4 (per-model intercept and slope) is indistinguishable from M0 (per-decision only), and allowing the model intercept and slope to correlate (M_corr) does not help; the estimated correlation is -0.37 (94% HDI $[-0.97, 0.35]$), so a model that complies more by default is not more reinforceable.

Under M8 the interaction dominates: $\sigma_{\text{int}} = 1.84$ (94% HDI $[1.26, 2.40]$), larger than the per-instruction spread ($\sigma_\beta = 0.43$), the per-model intercept ($\sigma_{\text{model}} = 0.66$), and the per-model slope ($\sigma_{t\text{-model}} = 0.43$); see Fig. 3. The additive components shrink once the interaction is in the model, because in additive specifications they absorb interaction variance. This also explains the halved σ_β relative to the conference study: with a single model, the model-by-instruction interaction has nowhere to go and collapses into the per-instruction effect, inflating it. Fig. 4 shows the interaction directly: the conference study's

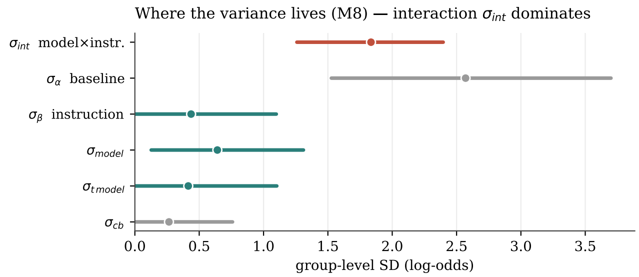

Fig. 3. Group-level standard deviations under M8. The model-by-instruction interaction σ_{int} is the largest source of structured variation; the additive per-model and per-codebase terms are small.

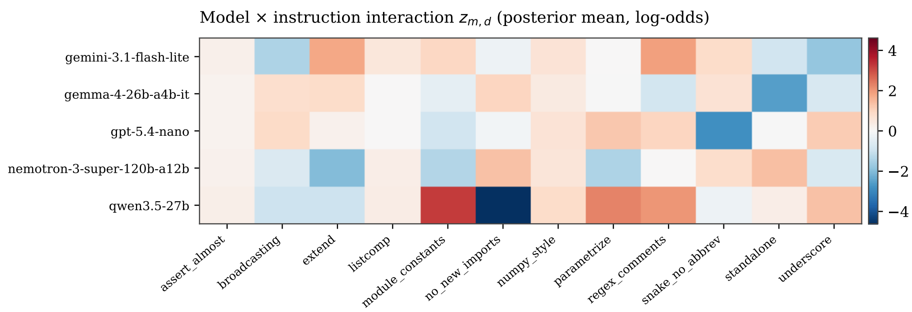

Fig. 4. Posterior mean of the model-by-instruction interaction $z_{m,d}$ (log-odds). Red raises, blue lowers compliance for that pair beyond the additive effects. Retention is a (model, instruction) property: which instruction is retained depends on the model.

primary model, Qwen, carries the most extreme cells, including the strongly negative `dependencies_no_new` cell that produced the original "backfire."

5.3 The reported backfire is a measurement artifact

The conference study reported that stating `dependencies_no_new` ("do not add new imports") *lowered* compliance, $\beta = -2.36$. Two independent analyses show this is not real. First, the original checker counted any import statement as new, including imports already present in the files shown to the model; being told "do not add imports" plausibly makes a model re-show existing imports, which the checker then penalized. With a context-aware checker that only counts genuinely new modules, the effect is null ($\beta = -0.11$ to -0.25 across specifications, interval spanning zero). Second, the framing experiment finds no effect of polarity: the pooled negation-minus-positive effect is 0.36 (94% HDI $[-0.13, 0.89]$), and every per-instruction interval crosses zero, so phrasing the rule as a negation does not reduce compliance. The artifact is instructive for measurement: under M_irt, `dependencies_no_new` has by far the highest discrimination ($a_d = 2.90$), i.e. it separates models most sharply, a reminder that a highly discriminating item can be measuring an artifact rather than the intended construct.

5.4 Variance components are pool-sensitive; rankings are not

The variance-component estimates depend on which models are pooled. Across the five-model, six-model, and ten-model sets, σ_β ranges 1.0–1.7 and σ_{model} ranges 0.48–0.98; the interaction conclusion and the per-instruction *rankings* are stable, but the absolute magnitudes of the variance components are not. A power-scaling sensitivity analysis confirms that the variance components are prior-influenced (a consequence of having few groups), while the fixed effects are not. We therefore report rankings and the interaction structure as robust, and treat the variance magnitudes as estimates with genuine prior dependence.

5.5 Measurement validity

Checker–judge agreement is weak: the quadratic-weighted κ between the deterministic checker and the cross-family judge panel has a panel median of 0.13, with individual judges from 0.03 to 0.17. Agreement is lowest on judgment-laden style and documentation decisions and higher on syntactically unambiguous ones, on which the robust effects sit. We read this as a limit on construct validity for the softer decisions, not as support for replacing the reproducible checker with a judge.

6 Discussion

The central result reframes the phenomenon. Per-instruction retention is not a property of the instruction alone, nor of the model alone, but of the (model, instruction) pair. An additive analysis, including the conference model and any report of per-instruction means, misattributes interaction variance to main effects and can manufacture artifacts like the "backfire," which was a single extreme cell of one model amplified by a checker bug. A reinforcement policy that follows from this is necessarily model-specific: there is no instruction-level ranking that transfers across models, because the ranking is itself a function of the model.

For deployed agents the implication is direct. A guardrail is an instruction, and whether it survives a long run depends on the specific (model, guardrail) pair, not on a generic decay rate. An operator cannot read off from one model which constraints another model will drop, and an aggregate compliance number hides the interaction entirely. Per-instruction, per-model monitoring is the prerequisite for noticing silent decay.

The methodological lesson generalizes beyond retention. Evaluation data is typically systems crossed with items, scored on graded rubrics, and the quantity of interest is often comparative. The model that fits this data well, a hierarchical Bayesian graded-response model with crossed effects and a model-by-instruction interaction, is exactly what the item-response and generalizability-theory traditions prescribe, yet current eval practice reports raw means or, at best, additive uncertainty. Our ladder shows the cost concretely: the additive models are 59 ELPD worse and miss the dominant structure. We suggest this model as a default for eval analysis and develop the argument in a companion paper.

7 Limitations

The variance components are prior-sensitive and pool-dependent (§5.4); we report rankings and structure rather than leaning on the magnitudes. The framing arms cover three of the five models, so the polarity null is not full-factorial. Compliance is measured at turns 20–25 only, a retention snapshot, not a decay curve; we do not estimate per-instruction forgetting rates. Checker–judge agreement is low on softer decisions (§5.5), a construct-validity limit. The selective reinforcement we motivate is not implemented as a closed-loop policy; we estimate the structure such a policy would exploit, not its closed-loop performance. Context length varies across codebases rather than being held constant, though the per-codebase variance is small.

8 Conclusion

Per-instruction retention in long coding conversations is governed by a model-by-instruction interaction: an interaction model fits about 59 ELPD better than additive models, additive per-model terms add nothing, and which instruction is retained depends on the model. The conference study's single-model design collapsed this interaction into an inflated per-instruction effect and, together with a checker artifact, produced a spurious "backfire" that does not replicate. The heterogeneity is real but smaller and differently shaped than first reported. Beyond the retention question, the analysis argues for a specific standard: evaluation data with systems crossed against graded items should be analyzed with a hierarchical Bayesian interaction model, not raw means.

Use of large language models. During the preparation of this work the author used a large language model (Claude) to improve the language and readability of the manuscript and to assist with analysis and figure code; all results were computed by the deterministic checkers and statistical models and verified by the author.

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